

Learning the Domain Before Writing Requirements

A Pilot Study on LLM-Mediated Understanding in Requirements Engineering

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Abstract

Large language models have increasingly been used to support software engineering tasks, including requirements elicitation, domain familiarization, business rule extraction, and the generation of structured requirements artifacts. However, many uses of these models still treat developers and analysts as homogeneous users, even though requirements engineering depends not only on information availability, but also on how unfamiliar domain knowledge is explained, organized, and transformed into analyzable rules. This paper reports an exploratory pilot study conducted in a graduate-level Master's course in Computer Science on Intelligent Software Engineering. The study investigates whether personality-aware prompting can support the initial learning of business rules in an unfamiliar domain. The Big Five/OCEAN model, operationalized through IPIP-NEO profiles, was used to generate personalized instructional prompts for developers working with business rules related to public works monitoring. Eight participants completed a pre-test, received LLM-generated instructional material under two experimental conditions, and completed post-tests about two domain artifacts: Measurement Bulletins and Construction Monitoring Reports. The results show a substantial increase from pre-test to post-test scores, with the average score rising from 31.82% to approximately 93.47%. At the condition level, the standard structured prompt reached 93.18%, while the personality-aware condition reached 93.75%, a small descriptive advantage of 0.57 percentage points. This difference is not sufficient to support a conclusive claim that personality-aware prompting outperformed the structured baseline. These findings suggest that LLM-generated material can act as an effective didactic mediator for initial domain learning, while also showing that stronger instruments are needed to evaluate whether personality-based adaptation produces meaningful additional gains. The paper discusses implications for Intelligent Software Engineering, especially the need for rigorous experimental protocols, stronger transfer tasks, expert validation, and evaluation of participant-produced requirements artifacts in future work.

Keywords

Intelligent Software Engineering, Requirements Engineering, Large Language Models, Personality-Aware Prompting, Big Five, IPIP-NEO, Business Rules, Domain Learning

1 Introduction

Requirements engineering frequently requires software professionals to understand domains that are outside their original technical expertise. In real projects, developers, analysts, and product

teams must transform fragmented domain explanations into entities, fields, validations, exceptions, business rules, and system behavior. This task is especially difficult when the domain involves institutional procedures, legal constraints, public administration, accountability, and specialized documents. In these contexts, the challenge is not only to produce a syntactically correct requirements artifact, but also to learn enough of the domain to avoid confusing planning documents with execution records, operational logs with managerial reports, or fiscal validation with technical monitoring. Recent advances in large language models (LLMs) have created new opportunities for supporting requirements engineering. LLMs can summarize domain documents, explain terminology, propose candidate requirements, identify ambiguities, and organize information into structured artifacts. These capabilities make them relevant to Intelligent Software Engineering (ISE), understood here as the use of artificial intelligence techniques to support software engineering activities. In requirements engineering, LLMs may work as assistants that help practitioners move from informal domain knowledge to more explicit representations, such as business rules, validation constraints, traceable data fields, and requirement structures. However, the use of LLMs in requirements engineering is often centered on the task and the prompt, not on the human learner. A typical prompt asks the model to generate requirements, extract rules, classify information, or explain a domain. Less attention is given to the fact that different practitioners may benefit from different forms of explanation. A senior developer may prefer concise and formal structures. A junior developer may need more examples and intermediate checkpoints. A participant with high openness may respond well to conceptual exploration, while a highly conscientious participant may prefer explicit criteria, constraints, and validation rules. In this sense, the effectiveness of LLM support may depend not only on what the model knows, but also on how the model mediates the domain for a specific person. This paper investigates this issue through an exploratory pilot study on personality-aware prompting for requirements learning. The study uses the Big Five/OCEAN model, operationalized through IPIP-NEO profiles, as a basis for adapting the didactic style of LLM-generated material. The domain selected for the study is the monitoring of public works executed indirectly, where an external contracted company performs the work while the public agency monitors, validates, and consolidates information about execution. The two focal artifacts are the Measurement Bulletin, which records measured execution for validation and payment, and the Construction Monitoring Report, which consolidates managerial and technical information about progress, risks, delays, and pending issues. The initial ambition of the study was to evaluate not only post-test

learning, but also the quality of requirements artifacts produced by participants after interacting with LLM-generated material. Due to operational constraints, limited time, and restricted institutional availability, the artifact production and expert evaluation stage could not be fully executed. Therefore, this paper reports the study as a pilot focused on protocol validation and initial domain learning, measured by pre-test and post-test instruments. This limitation is treated explicitly: the results do not support claims about improved requirements artifact quality. Instead, they support a narrower claim about the use of LLM-generated material as a didactic mediator for learning business-rule concepts in an unfamiliar domain. The contribution of this paper is threefold. First, it presents a reproducible pilot protocol for evaluating LLM-assisted domain learning in requirements engineering. Second, it provides preliminary empirical evidence that LLM-generated instructional material can substantially improve post-test performance in a specialized requirements domain. Third, it analyzes the small and non-conclusive descriptive advantage observed for the personality-aware condition and interprets this result as a methodological finding. In particular, the paper argues that personality-aware LLM support requires stronger evaluation instruments, transfer tasks, artifact-based assessment, and expert validation before claims of effectiveness can be made.

1.1 Research Context and Motivation

The study was conducted in the context of a graduate-level Master's course in Computer Science on Intelligent Software Engineering. The goal was not to build a production-ready requirements engineering tool, but to investigate how AI-based mediation could be evaluated in a controlled, small-scale setting. This educational and exploratory context shaped the methodological decisions of the study. Instead of evaluating LLMs only as generators of requirements, the study treated the model as a mediator between a controlled technical domain base and participants who had limited prior knowledge of that domain. The selected domain was public works monitoring. This domain was useful for the study because it contains documents that are conceptually close, but functionally different. For example, a Physical-Financial Schedule represents planned execution over time, while a Measurement Bulletin records what was actually measured in a specific period. A Daily Construction Report records daily events, while a Construction Monitoring Report consolidates information for managerial and technical decision-making. These distinctions are typical of requirements engineering problems: the analyst must identify document purposes, entities, data fields, rules, validations, exceptions, and dependencies among artifacts. The domain also contains business rules that are easy to state but difficult to contextualize. A measured quantity should not exceed the contracted quantity without formal justification. A measured item should be linked to the analytical budget spreadsheet. A report should compare planned and executed progress. Delays, risks, pending issues, and evidence should be represented in ways that support traceability and decision-making. These rules are not merely interface details; they express constraints that affect software behavior, data validation, workflow, and accountability. From an Intelligent Software Engineering perspective, this setting raises a relevant question: can an LLM help practitioners

learn enough about an unfamiliar domain to reason about requirements? If yes, does adapting the explanation to a participant profile produce measurable benefits? These questions are aligned with a broader shift in software engineering research, where AI systems are increasingly used not only to automate tasks, but also to support human reasoning, comprehension, and decision-making.

1.2 Scope and Contributions

This paper should be read as a pilot study rather than a complete validation of personality-aware prompting for requirements engineering. The study involved eight participants, two domain tasks, two prompting conditions, and pre/post knowledge tests. It did not complete the planned stage in which participants would produce their own requirements artifacts for blind expert evaluation. As a consequence, the results are limited to knowledge acquisition as measured by the tests, and not to the quality of independently produced requirements specifications. Within this scope, the study makes the following contributions:

- It proposes a protocol for comparing standard structured prompting and personality-aware prompting in a requirements engineering learning task.
- It operationalizes personality-aware prompting using IPIP-NEO/OCEAN profiles as didactic adaptation signals.
- It applies the protocol to a realistic domain involving business rules for public works monitoring.
- It reports pre-test and post-test results showing strong learning gains after interaction with LLM-generated material.
- It discusses the small and non-conclusive descriptive advantage observed for the personality-aware condition and interprets this result as a methodological finding.
- It identifies improvements needed for future Intelligent Software Engineering experiments, including stronger transfer tasks, expert-reviewed artifacts, and better instruments for detecting differences between conditions.

The condition-level comparison is central to the paper. Condition B obtained a small numerical advantage over Condition A, but the difference was only 0.57 percentage points. This does not support a strong claim that personality-aware prompting objectively outperformed the structured baseline. Rather, it suggests that both conditions produced high immediate learning scores and that the current experimental design was not sensitive enough to determine whether personality-aware adaptation provides a meaningful additional benefit. There are several possible explanations: the standard prompt may already have been sufficiently structured; the post-test may have been too close to the instructional material; the sample was small; and the test may have produced a ceiling effect. These threats are discussed later in the paper.

2 Background and Related Work

This section situates the study in three bodies of work: requirements engineering, large language models in software engineering, and personality-based adaptation. The goal is not to provide an exhaustive systematic review, but to identify the conceptual basis needed to understand the pilot. The study combines these areas by treating

the LLM not simply as a generator of candidate requirements, but as an intelligent mediator of domain learning for requirements work.

2.1 Requirements Engineering and Domain Learning

Requirements engineering is concerned with understanding stakeholder needs, domain constraints, system behavior, and the conditions under which software should operate. Traditional requirements engineering activities include elicitation, analysis, specification, validation, and management [1–3]. These activities depend heavily on domain understanding. Analysts must often distinguish between concepts that look similar from outside the domain but have different operational meanings inside the organization. In business-rule-intensive systems, requirements are not only user stories or functional descriptions. They include constraints, validation rules, exception handling, document dependencies, roles, approval flows, and conditions for state transitions. A requirements artifact may need to specify, for example, when a measured quantity becomes invalid, which document provides authoritative values, which role can approve a record, and what exception should be raised when the data violates a contractual rule. In this context, learning the domain is a precondition for producing useful requirements. The domain used in this study illustrates this problem. Public works monitoring involves multiple documents with different purposes. A contract defines object, value, deadline, and general conditions. An analytical budget spreadsheet provides items, quantities, unit prices, and total values. A physical-financial schedule represents planned progress over time. A measurement bulletin records executed and measured services. A daily construction report registers daily activities, occurrences, and evidence. A construction monitoring report consolidates the status of the work, including delays, risks, pending issues, and deviations. A developer unfamiliar with the domain may easily confuse these artifacts, producing requirements that are technically structured but semantically wrong. Thus, the central problem addressed by this pilot is not only requirement generation. It is domain learning for requirements reasoning. The study asks whether LLM-generated instructional material can help participants acquire enough conceptual clarity to answer questions about business rules and document relationships. This learning-oriented framing is important because it differs from evaluations in which the LLM output itself is treated as the final requirements artifact.

2.2 Large Language Models in Software Engineering

LLMs have been explored in many software engineering tasks, including code generation, test generation, bug fixing, program comprehension, documentation, and requirements engineering [4, 5]. In requirements engineering, LLMs may be used to summarize stakeholder input, generate user stories, identify inconsistencies, propose acceptance criteria, extract candidate requirements from text, or transform informal descriptions into structured specifications. These applications are attractive because requirements work is language-intensive and often depends on interpreting ambiguous information. Nevertheless, LLM use in requirements engineering

raises important concerns. Models may produce plausible but incorrect statements, invent constraints, omit important exceptions, or overstate confidence. In high-stakes domains, such as public administration, health, finance, or infrastructure, these risks make it insufficient to evaluate whether an output is fluent. The output must be validated against domain knowledge and expert judgment. For this reason, this study used a controlled technical base as input to the LLM and treated the generated material as an instructional artifact rather than as an authoritative specification. The use of LLMs in Intelligent Software Engineering should therefore be understood as a socio-technical arrangement. The model is not an autonomous engineer. It is part of a workflow that includes domain sources, prompt design, human interpretation, validation, and revision. In this paper, the LLM plays the role of didactic mediator: it receives a controlled domain base and produces instructional material intended to help participants reason about requirements. The evaluation focuses on whether participants improve their domain knowledge after interacting with this material. Prompt design is a key factor in this process. Prior work on prompt patterns has shown that prompts can be structured to control output format, reasoning behavior, constraints, and interaction style [5]. In requirements engineering, prompt structure can be used to request entities, fields, business rules, validations, exceptions, ambiguities, and document relations. This pilot builds on that idea but adds a human-centered dimension: the possibility that the same technical content can be presented differently depending on the participant profile.

2.3 Personality-Aware Adaptation and IPIP-NEO

The Big Five model is one of the most widely used frameworks for describing personality traits. It organizes personality into five broad dimensions: openness, conscientiousness, extraversion, agreeableness, and neuroticism [6, 7]. The IPIP project provides public-domain personality items that can be used to operationalize these traits and their facets [8, 9]. In this study, IPIP-NEO/OCEAN profiles were not used to classify participants in a clinical sense, nor to judge their capabilities. They were used only as didactic adaptation signals. The assumption behind personality-aware prompting is modest. Different people may prefer different instructional styles. A highly conscientious participant may benefit from explicit structure, criteria, and validation steps. A participant with high openness may benefit from conceptual exploration and examples. A participant with higher neuroticism may benefit from reduced overload and clearer checkpoints. A participant with higher extraversion may engage better with collaborative and action-oriented explanations. These are not deterministic claims. They are hypotheses about how explanation style might be adapted. In the present study, each participant had an anonymized profile summarized into a didactic strategy. The prompt instructed the LLM to use the profile only to adapt form, not technical content. It also instructed the LLM not to mention that a psychological profile was being used, not to label the participant, and not to make evaluative statements about personality. This constraint was important for ethical and methodological reasons. The purpose was to test whether adaptation of explanation style could improve learning, not to expose or interpret participant personality. This design connects personality-aware

adaptation with Intelligent Software Engineering. If software engineering tools are increasingly AI-mediated, future tools may need to adapt not only to code context, project context, and domain context, but also to the human user. However, this pilot also shows that such claims require careful evaluation. A personalized prompt may appear intuitively useful, but without sensitive instruments and expert-validated outcomes, it is difficult to demonstrate measurable benefit.

3 Method

This section describes the pilot study design. The study was conducted as part of a graduate-level Master’s course in Computer Science on Intelligent Software Engineering. The aim was to evaluate a small but reproducible protocol for LLM-mediated domain learning in requirements engineering. The protocol included a pre-test, personality profiling, controlled domain material, LLM-generated instructional materials under two conditions, and post-tests. The planned evaluation of participant-produced artifacts was not executed due to operational constraints and is therefore treated as future work.

3.1 Research Questions

The pilot was guided by three research questions: **RQ1.** Does interaction with LLM-generated instructional material improve participants’ knowledge of business rules in an unfamiliar requirements domain? **RQ2.** Does personality-aware prompting lead to higher post-test performance than standard structured prompting? **RQ3.** What methodological lessons does the pilot provide for future Intelligent Software Engineering studies involving LLM-mediated requirements learning? RQ1 focuses on the overall effect of the intervention, regardless of condition. RQ2 focuses on the comparison between the standard and personalized prompt conditions. RQ3 recognizes the exploratory nature of the study and treats limitations as useful evidence for improving future protocols.

3.2 Participants and Anonymization

Eight participants were included in the study. They were anonymized as P01 to P08. The group was intentionally heterogeneous in professional experience, including senior developers, analysts, support professionals, interns, and requirements-oriented practitioners. The purpose was not statistical generalization, but exploratory observation of how participants with different backgrounds responded to LLM-generated instructional materials. Table 1 summarizes the anonymized participants and the experimental assignment. The names used internally during data collection were replaced by participant codes in the analysis. The public repository contains only anonymized material and does not expose raw personality test responses. The design balanced task and condition. Four participants received the Measurement Bulletin under Condition A and four under Condition B. Similarly, four participants received the Construction Monitoring Report under Condition A and four under Condition B. This balance was intended to reduce the risk that one condition would be favored simply because it was associated with an easier task.

3.3 Domain and Learning Tasks

The domain concerned public works executed indirectly. In this arrangement, a contracted company performs the construction work, while a public agency monitors execution, validates information, and consolidates records. The domain was selected because it contains document relationships that are conceptually rich and relevant to requirements engineering. Two learning tasks were used. The first task involved the Measurement Bulletin. Participants had to understand that this artifact records measured execution in a period, supports validation, and may provide a basis for payment. They also had to understand its relationship with the contract, analytical budget spreadsheet, physical-financial schedule, evidence, and approval workflow. The second task involved the Construction Monitoring Report. Participants had to understand that this artifact consolidates information about the general status of the work. It does not replace the Measurement Bulletin or the Daily Construction Report. Instead, it consumes information from them and from other documents to support managerial and technical monitoring. Participants had to reason about progress, delays, risks, pending issues, evidence, and planned versus executed comparison.

3.4 Controlled Technical Base

A key methodological decision was to provide the LLM with a controlled technical base. The goal was not to test whether the model already knew the domain from pretraining. Instead, the model was supplied with a prepared domain explanation that included concepts, documents, rules, validations, exceptions, and ambiguities. This technical base was identical across conditions. The base described the following artifacts: contract, analytical budget spreadsheet, physical-financial schedule, daily construction report, measurement bulletin, and construction monitoring report. It also included business rules such as linking a measurement bulletin to a contract, validating measured items against the budget spreadsheet, comparing planned and executed progress, and generating alerts for divergences or missing evidence. Ambiguities requiring expert validation were also included, such as who can approve records, which evidence is mandatory, how to treat complementary measurements, and what tolerance should be used for delays. This design makes the experiment more controlled. The difference between conditions was not the domain knowledge available to the LLM. The difference was the instructional strategy used to present that knowledge to participants.

3.5 Experimental Conditions

The study compared two conditions. **Condition A: Standard structured prompting.** The LLM received the technical domain base, the task, and a structured output format. It was instructed to separate concepts, rules, requirements, validations, exceptions, document relations, and ambiguities. It was also instructed not to invent specific legislation and to indicate points requiring specialist validation. **Condition B: Personality-aware prompting.** The LLM received the same technical domain base and task, but also an anonymized OCEAN/IPIP-NEO didactic profile. The prompt instructed the model to adapt the explanation style according to the profile while keeping the technical content unchanged. It also instructed the model not to mention the psychological profile, not

Table 1: Participants and experimental assignment

Code	Professional profile	Assignment
P01	Senior public-sector developer	A: Bulletin; B: Report
P02	Software development scholarship holder	B: Bulletin; A: Report
P03	Senior IT analyst and data scientist	A: Bulletin; B: Report
P04	General IT coordinator	B: Bulletin; A: Report
P05	Level-2 support analyst	A: Bulletin; B: Report
P06	Development intern	A: Report; B: Bulletin
P07	IT developer with mathematics background	A: Bulletin; B: Report
P08	Requirements developer	A: Report; B: Bulletin

to label the participant, and not to alter business rules based on personality. The intended contrast was therefore not between more information and less information. Both conditions had access to the same technical content. The contrast was between a generic structured explanation and a profile-adapted explanation.

3.6 Instruments

The study used three main instruments: a pre-test, LLM-generated instructional materials, and post-tests. The pre-test, referred to as K0, measured initial knowledge of the domain. It included questions about the distinction between planned and executed information, the purpose of measurement documents, monitoring reports, delays, evidence, responsibilities, and validation rules. The maximum score was 44 points. The K0 instrument was broader than the task-specific K1 instruments. It measured initial familiarity with the public works monitoring domain, including distinctions between planning and execution, the role of measurement documents, the purpose of monitoring reports, evidence, responsibilities, delays, and validation rules. The K1 instruments were narrower and task-specific, focusing on whether participants understood the artifact associated with the instructional material they had just received. Both were reported as percentages to support a descriptive pre/post comparison. The post-tests, referred to as K1, were applied after exposure to the instructional materials. There were two post-test versions: one for the Measurement Bulletin and one for the Construction Monitoring Report. Each post-test had 11 knowledge questions, scored from 0 to 2, and one self-confidence question from 1 to 5. The self-confidence question was not included in the knowledge score. The K1 scoring rubric was:

- 0: incorrect, absent, or outside the domain;
- 1: partially correct, generic, or containing relevant imprecision;
- 2: correct and adequate.

Thus, each task-specific K1 had a maximum of 22 points. Since each participant completed two K1 tests, the total post-test score per participant was 44 points.

3.7 Procedure

The study followed a sequential procedure. First, participants completed the K0 pre-test. Then, IPIP-NEO/OCEAN profiles were obtained and converted into anonymized didactic summaries. Next, the controlled domain base was prepared and used to generate

instructional materials via API calls to the LLM. Each call was independent and associated with a participant, condition, and task. The model, temperature, output format, and technical domain base were kept constant. Participants then received two materials in sequence, each associated with one task and one condition. The labels A and B were not presented as experimental labels to participants. After each material, participants answered the corresponding K1. The originally planned next step was to ask participants to produce their own requirements artifacts after reading each material. These artifacts would then be evaluated by specialists. This step was not completed and is therefore excluded from the results.

3.8 Study Workflow

Figure 1 summarizes the complete workflow of the pilot study. The process starts with the participant baseline, in which eight participants were identified by anonymized codes from P01 to P08 and completed the K0 pre-test. This first stage measured their initial domain knowledge about public works monitoring. In parallel, IPIP-NEO/OCEAN information was collected and transformed into anonymized didactic strategy summaries, which were used only in Condition B.

The second stage corresponds to material generation. Both experimental conditions received the same controlled technical domain base, including information about contracts, budget spreadsheets, physical-financial schedules, measurement bulletins, daily reports, and monitoring reports. Condition A used a standard structured prompt, while Condition B used the same structured prompt plus the anonymized didactic strategy. The materials were generated through scripted OpenAI API calls, with independent executions, no reused conversation history, and store=false. Thus, the intended experimental contrast was restricted to the explanatory form, while the technical content remained constant across conditions.

The third stage corresponds to intervention and evaluation. Participants received the generated instructional material and then completed K1 post-tests about the Measurement Bulletin and the Construction Monitoring Report. The evaluation compared the average K1 performance, the condition-level results, and the overall gain from K0 to K1. The figure also indicates the stage that was originally planned but not executed: participant-produced requirements artifacts followed by blind expert review.

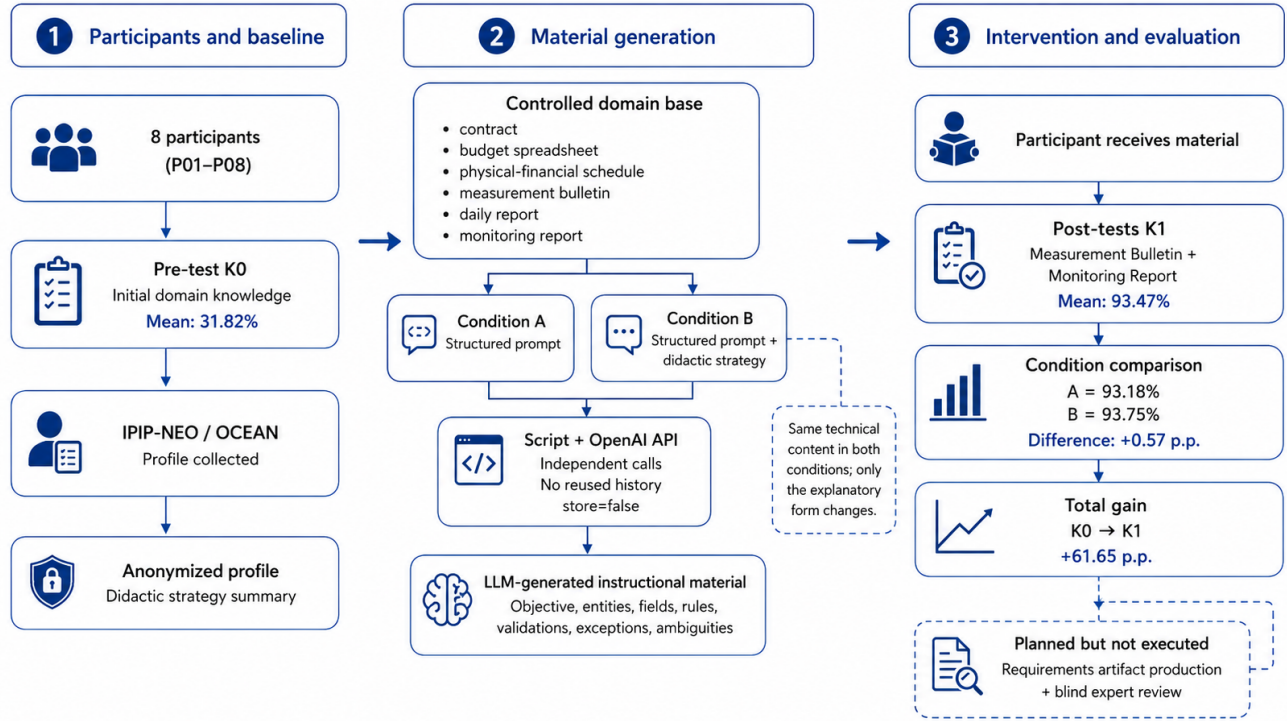


Figure 1: Workflow of the pilot study, from participant baseline and profile collection to LLM-based material generation, post-test evaluation, and the planned but non-executed artifact review stage.

3.9 LLM Execution and Contamination Control

The instructional materials were generated through scripted API calls rather than manually through an interactive chat interface. This decision was adopted to reduce manual variation and improve traceability. Each participant-task-condition combination was generated through an independent call. The script did not reuse conversation history, previous answers, or participant responses as input for later generations. Each prompt was self-contained and included the controlled technical domain base, the participant task, the mandatory output format, and the condition-specific instructions. In Condition A, no participant profile was included. In Condition B, the only additional input was the anonymized didactic strategy associated with the participant. Thus, the experimental manipulation was restricted to the explanatory style, while the technical content, task structure, and output structure were kept constant. The model, temperature, output format, and domain base were kept constant during the generation round. The script saved the exact prompt sent to the model, the generated response, and execution metadata such as participant code, condition, task, timestamp, model, temperature, response identifier, and token usage when available. The API call was configured with `store=false`. These measures were intended to reduce cross-condition contamination, preserve reproducibility, and support later inspection of the generated materials. A preliminary single-call execution was performed to estimate API cost before running the full generation

round. This preliminary call was not treated as a separate experimental condition.

3.10 Didactic Strategies Used in Condition B

In Condition B, the participant profile was converted into a short didactic strategy used by the prompt to adapt the form of the explanation. These strategies did not change the technical domain content. They only guided how the instructional material should be organized and presented. The strategies used in the prompt were the following:

- **P01:** use an objective, structured, and criteria-based explanation; avoid rhetorical excess; emphasize rules, validations, and logical order.
- **P02:** use a welcoming explanation organized into short steps; reduce overload; provide examples before abstraction; use understanding checkpoints.
- **P03:** use technical, direct, and logical language; avoid motivational tone; focus on structure, precision, dependencies, and exceptions.
- **P04:** use a decision-oriented and management-oriented explanation; emphasize practical impact; organize the material into short blocks.
- **P05:** use a practical, stable, and operational explanation; avoid excessive abstraction; provide concrete examples of fields, rules, and validations.

- **P06:** use a dynamic explanation with examples and checkpoints; break the task into small steps; avoid too many open options at the same time.
- **P07:** use a formal, logical, and well-delimited explanation; emphasize rules, consistency, and criteria; avoid vague analogies.
- **P08:** use a collaborative, structured, and requirements-oriented explanation; emphasize consensus, responsible roles, validation, and traceability.

In addition, the prompt explicitly instructed the model to adapt only the explanatory form, not the technical content. It also instructed the model not to mention the psychological profile, not to label the participant, and not to make evaluative statements about the participant.

4 Results

This section reports the pre-test and post-test results. The analysis is intentionally descriptive because the sample is small and the study is exploratory. The goal is not to claim statistical generalization, but to examine whether the protocol produced observable learning gains and whether the two prompting conditions differed in post-test performance. The main finding is that participants improved substantially from K0 to K1. At the condition level, Condition B obtained a small numerical advantage over Condition A, but the difference was too small to support a conclusive claim of superiority for personality-aware prompting.

4.1 Pre-Test Results

The K0 pre-test showed that the participants had limited and uneven prior knowledge of the domain. The average score was 14.00 out of 44 points, corresponding to 31.82%. This result was consistent with the expectation that the domain would be unfamiliar to several participants. Table 2 presents the anonymized K0 scores. The

Table 2: K0 pre-test baseline scores

Participant	K0 score	Percentage
P01	15/44	34.09%
P02	10/44	22.73%
P03	17/44	38.64%
P04	14/44	31.82%
P05	15/44	34.09%
P06	8/44	18.18%
P07	19/44	43.18%
P08	14/44	31.82%
Mean	14.00/44	31.82%

lowest initial score was observed for P06, followed by P02. These participants had more difficulty with domain distinctions before the intervention. The highest initial score was observed for P07. This variation is important because it suggests that the intervention was applied to a group with different levels of initial readiness. Qualitatively, the K0 responses indicated gaps in the distinction between planning and execution, uncertainty about the role of the Measurement Bulletin, and limited clarity about how monitoring reports

consolidate information from operational documents. These were precisely the distinctions targeted by the instructional materials.

4.2 Post-Test Results

The K1 results showed a strong increase in performance, but with variation across participants and tasks. The average K1 score was 41.13 out of 44 points, corresponding to 93.47%. Table 3 presents the anonymized post-test scores by document. The Measurement Bulletin scores were slightly higher on average, reaching 165 out of 176 possible points, or 93.75%. The Construction Monitoring Report scores reached 164 out of 176 possible points, or 93.18%. This difference is small, but it indicates that the two artifacts did not produce identical post-test performance. The Bulletin answers were generally strong. Participants correctly identified the Measurement Bulletin as an artifact for measured execution, distinguished it from the Physical-Financial Schedule, linked it to the Analytical Budget Spreadsheet, explained the need for contract linkage, and identified glosa as partial rejection or deduction of a measurement. Partial scores were mainly associated with ambiguities that required specialist validation. The Monitoring Report answers were also strong. Participants correctly stated that the report consolidates information and does not replace the Measurement Bulletin or the Daily Construction Report. They identified relevant input documents and explained the importance of comparing planned and executed progress. Partial scores were assigned mainly to answers that proposed fixed thresholds or narrowed the meaning of pending issues, both of which would require expert validation before becoming system rules.

4.3 Learning Gain from K0 to K1

The average score increased from 31.82% in K0 to 93.47% in K1. This corresponds to an average gain of 61.65 percentage points. Table 4 shows individual gains using the corrected K0 and K1 scores. The largest gains occurred among participants with lower prior knowledge, especially P02 and P06. This suggests that the LLM-generated materials were particularly useful for participants who initially lacked domain familiarity. In Intelligent Software Engineering terms, this result supports the idea that LLM-based assistants may help reduce the initial domain-understanding gap in requirements work. However, this gain should not be interpreted as proof that the LLM was superior to other instructional approaches. Since the study did not include a non-LLM instructional control group, the result indicates that the generated materials were associated with improved post-test performance, but it does not isolate the LLM as the only causal factor. A stronger future design should include questions involving new cases, conflicting evidence, incomplete documents, and ambiguous stakeholder statements.

4.4 Comparison Between Conditions A and B

The comparison between conditions showed high performance in both groups, with a small descriptive advantage for the personality-aware condition. Because the design was balanced, each condition included four Measurement Bulletin post-tests and four Construction Monitoring Report post-tests. Condition A reached 164 out of 176 possible points, corresponding to 93.18%. Condition B reached

Table 3: K1 post-test scores by document

Participant	Bulletin	Report	Total	Percentage
P01	22/22	22/22	44/44	100.00%
P02	21/22	19/22	40/44	90.91%
P03	20/22	21/22	41/44	93.18%
P04	21/22	19/22	40/44	90.91%
P05	22/22	21/22	43/44	97.73%
P06	19/22	18/22	37/44	84.09%
P07	22/22	22/22	44/44	100.00%
P08	18/22	22/22	40/44	90.91%
Mean	20.63/22	20.50/22	41.13/44	93.47%

Table 4: Learning gain from K0 to K1

Participant	K0	K1	Gain
P01	34.09%	100.00%	+65.91 p.p.
P02	22.73%	90.91%	+68.18 p.p.
P03	38.64%	93.18%	+54.54 p.p.
P04	31.82%	90.91%	+59.09 p.p.
P05	34.09%	97.73%	+63.64 p.p.
P06	18.18%	84.09%	+65.91 p.p.
P07	43.18%	100.00%	+56.82 p.p.
P08	31.82%	90.91%	+59.09 p.p.
Mean	31.82%	93.47%	+61.65 p.p.

165 out of 176 possible points, corresponding to 93.75%. Table 5 summarizes the corrected condition-level result. This result answers

Table 5: Corrected condition-level K1 performance

Condition	K1 performance	Difference from A
A: Standard structured	164/176 = 93.18%	–
B: Personality-aware	165/176 = 93.75%	+0.57 p.p.

RQ2 cautiously. The personality-aware condition obtained a small numerical advantage, but the difference is not large enough to support a strong claim that personality-aware prompting outperformed the standard structured prompt in this pilot. The most appropriate interpretation is that both conditions produced high immediate learning scores, while the current design was not sensitive enough to establish a meaningful effect of personality-based adaptation. Several factors may explain this result. The standard prompt was already explicit and well structured. The controlled technical base was sufficiently informative in both conditions. The post-test focused on recall and near-transfer. In addition, the overall scores were close to the maximum, which may have reduced the ability of the instrument to distinguish between prompting strategies.

4.5 Self-Reported Confidence

All participants reported confidence level 4 out of 5 after each post-test. This result indicates that participants felt reasonably confident

after the intervention, but it also shows that the confidence measure had low discriminatory power. Since all participants selected the same level, the measure could not distinguish between conditions or participant profiles. In future work, confidence should be measured with more sensitive instruments. Instead of a single item, the study could include perceived clarity, perceived cognitive load, perceived usefulness, perceived effort, trust in the material, and confidence in producing an artifact independently. These measures may be more appropriate for detecting the effect of personality-aware explanation style, even when objective knowledge scores are similar.

5 Discussion

The results should be interpreted as evidence from a pilot study, not as definitive validation of personality-aware prompting. The main finding is twofold. First, LLM-generated instructional materials were associated with a substantial increase in domain knowledge from pre-test to post-test. Second, the personality-aware condition obtained only a small descriptive advantage over the standard structured condition in objective K1 scores. This combination is important for Intelligent Software Engineering because it separates two claims that are often mixed: LLMs may be useful as domain-learning mediators, but personality-based adaptation requires stronger evidence before it can be claimed as an effective improvement over a well-structured baseline.

Table 6: Research questions, evidence, and interpretation

Research question	Evidence	Interpretation
RQ1: Does LLM-generated material support domain learning?	K0 average increased from 31.82% to 93.47% in K1.	The material was associated with substantial immediate domain learning, although the design does not isolate the LLM as the only causal factor.
RQ2: Does personality-aware prompting improve post-test performance?	Condition A reached 93.18%, while Condition B reached 93.75%.	Condition B had a small descriptive advantage of 0.57 p.p., but the difference is not conclusive.
RQ3: What methodological lessons emerge?	High K1 scores, low condition difference, confidence homogeneity, and non-executed artifact review stage.	Future studies need stronger transfer tasks, expert-reviewed artifacts, richer perception measures, and larger samples.

5.1 LLM as a Didactic Mediator in Requirements Engineering

The strongest result is the increase from K0 to K1. Participants who initially had limited knowledge of public works monitoring were able to answer post-test questions about document purposes, business rules, validations, evidence, and document relationships. This suggests that LLM-generated material, when grounded in a controlled technical base, can support rapid domain familiarization. In requirements engineering, this is relevant because domain familiarization is often a bottleneck. Practitioners need to understand what documents mean, how they relate to each other, and which constraints affect system behavior. The LLM can help organize this knowledge into a structure that is closer to requirements work: objective, entities, fields, rules, validations, exceptions, and ambiguities. This organization is useful because it bridges the gap between domain explanation and requirements specification. However, the study does not demonstrate that the LLM improved the quality of requirements artifacts produced by participants. That stage was not executed. Therefore, the proper interpretation is that the LLM supported initial learning as measured by post-tests. The next step is to verify whether this learning translates into better requirements artifacts.

5.2 Why the Personality-Aware Advantage Was Not Conclusive

The small difference between conditions is one of the most important findings. Condition B reached 93.75%, while Condition A reached 93.18%, resulting in a descriptive advantage of 0.57 percentage points for the personality-aware condition. However, this difference is too small to justify a strong conclusion that personality-aware prompting was more effective than the structured baseline. There are at least four possible explanations. First, the baseline was strong. Condition A was not an unstructured prompt. It already instructed the model to separate concepts, rules, validations, exceptions, and ambiguities. This may have made the material sufficiently clear for most participants. Second, the controlled technical base was rich. Since both conditions received the same domain base, the model had enough information to generate high-quality instructional material regardless of personalization. Third, the K1 instrument may have produced a ceiling effect. Scores were very

high, leaving little room for the instrument to capture differences between conditions. Fourth, the post-test measured immediate understanding, not deeper transfer. Personality-aware prompting may affect engagement, perceived clarity, comfort, motivation, or ability to produce independent artifacts. These effects may not appear clearly in a short knowledge test. Thus, the result does not invalidate personality-aware prompting as a research direction. It clarifies that future studies need more sensitive instruments, stronger transfer tasks, and artifact-level evaluation before claiming that personality-based adaptation produces meaningful improvements.

5.3 Implications for Intelligent Software Engineering

The study has several implications for Intelligent Software Engineering. First, intelligent software engineering tools should not be evaluated only by the quality of generated text. In requirements engineering, the human practitioner remains central. A tool may be valuable because it helps the practitioner understand the domain, ask better questions, identify ambiguities, or produce better artifacts. Second, personalization must be evaluated carefully. It is tempting to assume that adapting prompts to user profiles will improve outcomes. This study shows that such an assumption is not enough. Even when the personality-aware condition obtains a small numerical advantage, the difference may be too small to support a strong claim if the baseline is already strong and the evaluation instrument is not sensitive. Third, LLM-based requirements tools should be grounded in controlled domain bases whenever possible. Without grounding, it becomes unclear whether the model is teaching controlled domain knowledge or producing plausible statements from pretraining. The controlled base used in this pilot was essential for methodological clarity. Fourth, future ISE systems should combine LLM generation with expert validation and traceable artifacts. Requirements engineering is not only about generating candidate rules; it is about producing accountable, validated, and maintainable specifications. The LLM can support this process, but should not replace specialist judgment. For ISE, this suggests that the unit of evaluation should not be only the generated artifact, but the human-AI learning process that precedes requirements production.

5.4 Methodological Lessons

The pilot provides methodological lessons for future work. The first lesson is that pre/post tests are useful but insufficient. They can measure immediate learning, but not necessarily artifact quality or transfer. The second lesson is that post-tests should include cases that differ from the instructional material. Otherwise, participants may reproduce language and examples rather than demonstrate independent reasoning. The third lesson is that expert evaluation should be planned as a central stage, not as an optional addition. In requirements engineering, specialists are needed to judge whether rules are correct, dangerous, incomplete, or institution-specific. The fourth lesson is that response homogeneity should be treated as data. In this study, many answers repeated similar expressions, such as fixed delay thresholds and decimal rounding ambiguities. This suggests that the material strongly shaped participant responses. That may be positive for teaching core concepts, but negative for evaluating deeper understanding.

6 Threats to Validity

This section discusses the main threats to validity. Because the study is exploratory and small-scale, these threats are substantial. Making them explicit is important to avoid overclaiming and to guide future iterations of the protocol.

6.1 Internal Validity

The main internal threat is the absence of a no-intervention control group. The study compares pre-test and post-test performance after exposure to LLM-generated material, but it does not compare against participants who received no material or non-LLM material. Therefore, the gain cannot be attributed exclusively to the LLM as a technology. It may reflect the effect of receiving any well-structured instructional material. Another internal threat is the proximity between instructional material and post-test questions. Since the K1 questions were aligned with the material, high performance may reflect near recall rather than robust transfer. Future studies should include transfer tasks involving new documents, conflicting information, and incomplete requirements scenarios.

6.2 Construct Validity

The study measured knowledge through short-answer and multiple-choice questions. This captures some aspects of domain understanding, but not the full construct of requirements engineering competence. A participant may answer conceptual questions correctly and still struggle to produce a high-quality requirements artifact. The personality-aware condition also raises construct validity issues. The study used summarized OCEAN/IPIP-NEO profiles as prompt inputs. This operationalization may be too coarse to capture meaningful differences in learning style. Moreover, the adaptation was produced by the LLM, and the study did not independently measure whether the generated explanations actually differed in ways aligned with the profiles.

6.3 External Validity

The sample included only eight participants and was conducted in the context of a graduate course. The findings cannot be generalized to all software professionals, all requirements engineering tasks,

or all domains. The domain of public works monitoring is also specific and document-heavy. Results may differ in domains such as healthcare, banking, education, or embedded systems. Nevertheless, the setting has ecological value because many real software projects require developers to learn unfamiliar institutional domains. The pilot should therefore be seen as a basis for future studies, not as final evidence.

6.4 Conclusion Validity

No inferential statistical analysis was performed because of the small sample and the exploratory nature of the study. The comparison between conditions is descriptive. The corrected condition-level results show a small numerical advantage for Condition B, but this difference is not sufficient to support a conclusive claim that personality-aware prompting improved objective learning outcomes. Larger samples, more sensitive instruments, and artifact-level evaluation are needed.

7 Artifact Availability

Reproducibility is especially important in studies involving LLMs because prompts, model settings, domain bases, and scoring decisions can strongly affect results. For this reason, the study was designed to be accompanied by a public repository containing anonymized artifacts, scripts, generated materials, and methodological documentation. The repository is intended to include:

- the Python script used to generate LLM materials via API;
- prompt construction logic for Conditions A and B;
- anonymized generated prompts and responses;
- documentation of the experimental protocol;
- documentation of the scoring procedure;
- corrected aggregate results;
- README documentation describing how to run the script.

The repository URL is:

<https://github.com/jaonunesunb/personality-aware-prompting>

Before publication, API keys, personal identifiers, raw psychological test details, and any sensitive institutional information must be removed. Only anonymized scores, didactic strategy summaries, prompts, and generated materials should be made public. For the public artifact package, the profiles can be reduced to didactic strategies and raw trait descriptions are not required for reproduction.

8 Ethics and Privacy

The study used personality information only as an instructional adaptation signal. The profiles were not used to evaluate participants, classify competence, diagnose behavior, or make decisions about individuals. In the prompt, the LLM was instructed not to mention the psychological profile, not to label the participant, and not to alter technical content based on personality. The public artifacts should preserve participant anonymity. Names used during data collection must be replaced by participant codes. The repository should not expose raw IPIP-NEO responses or identifiable personal information. The ethical position of this pilot is that personality-aware support in software engineering must be designed carefully: adaptation may be useful, but it can also become invasive or misleading if profiles are overinterpreted.

9 Conclusion

This paper presented an exploratory pilot study on personality-aware prompting for requirements engineering learning in the context of Intelligent Software Engineering. The study investigated whether LLM-generated instructional materials could help participants learn business-rule concepts in an unfamiliar domain and whether adapting the material to OCEAN/IPIP-NEO profiles would improve post-test performance. The results showed a substantial increase from pre-test to post-test scores. The average score rose from 31.82% to approximately 93.47%, suggesting that the LLM-generated materials were effective as didactic mediators for initial domain learning. The largest gains occurred among participants with lower initial knowledge, indicating that LLM-based support may be especially useful for reducing early domain-understanding gaps. The personality-aware condition obtained a small descriptive advantage over the standard structured condition. Condition A reached 93.18%, while Condition B reached 93.75%. However, the difference of 0.57 percentage points is not sufficient to support a conclusive claim that personality-aware prompting improved objective knowledge acquisition in the tested setting. Instead, the pilot shows that both a strong structured prompt and a personality-aware prompt grounded in the same controlled technical base can produce high immediate learning scores. The study also clarified important limitations. The planned artifact production and expert evaluation stage was not completed. As a result, the study cannot claim that participants produced better requirements artifacts or that personality-aware prompting improves requirements quality. Future work should complete this stage, include blind expert review, use transfer tasks, measure cognitive load and perceived clarity, increase the sample size, and explore additional conditions such as retrieval-augmented or knowledge-graph-based prompting. In summary, the study contributes a pilot protocol and preliminary evidence for LLM-mediated domain learning in requirements engineering. Its main result is not that personality-aware prompting is proven effective, but that a small descriptive advantage is insufficient without careful experimental design, sensitive instruments, and artifact-level validation. This is a useful lesson for Intelligent Software Engineering: intelligent assistance should be measured not only by fluent outputs, but by how well it supports human understanding, validation, and responsible software specification.

A Executed Protocol Checklist

This appendix summarizes what was executed in the pilot and what remains for future work. It is included to make the scope of the study explicit. The non-executed stages are not treated as results. They are part of the planned continuation of the protocol.

B Post-Test Structure

The K1 instrument had two versions. The Measurement Bulletin version asked participants to explain the purpose of the bulletin, distinguish it from the Physical-Financial Schedule, identify its main representation as measured execution, list essential fields, justify contract linkage, explain what happens when measured quantity exceeds contracted quantity, cite validations, identify the Analytical Budget Spreadsheet as the source for items and prices, describe evidence, define glosa, and cite an ambiguity requiring specialist

Table 7: Executed and non-executed protocol stages

Stage	Status
K0 pre-test	Executed
IPIP-NEO/OCEAN profiling	Executed
Profile consistency check	Executed
Controlled technical domain base	Executed
Prompt templates for A and B	Executed
LLM material generation via API	Executed
K1 post-tests	Executed
Participant-produced artifacts	Not executed
Blind expert evaluation of artifacts	Not executed
Knowledge graph condition	Future work

validation. The Construction Monitoring Report version asked participants to explain the purpose of the report, clarify that it does not replace the Measurement Bulletin or the Daily Construction Report, identify its main representation as managerial and technical consolidation, list essential fields, identify supporting documents, explain planned-versus-executed comparison, cite rules for delay or risk, explain pending issues, describe the role of evidence, distinguish occurrence, pending issue, and risk, and cite an ambiguity requiring specialist validation. The final item in each version asked participants to report confidence from 1 to 5. This item was analyzed separately and not included in the knowledge score.

C Prompt Design Summary

Condition A used a structured prompt. It instructed the LLM to act as a requirements engineering assistant, use the controlled domain base, and produce instructional material organized into objective, entities, fields, business rules, validations, exceptions, document relations, ambiguities, and final observations. Condition B used the same prompt structure and the same technical domain base. The only additional input was an anonymized didactic profile derived from OCEAN/IPIP-NEO. The LLM was instructed to adapt the form of explanation, not the technical content. It was also instructed not to mention the psychological profile and not to judge the participant. This design was intended to isolate didactic adaptation as the experimental variable.

D Future Expert Review Form

The next iteration of the study should include participant-produced artifacts. Specialists should evaluate them using a rubric from 0 to 3:

- 0: absent or incorrect;
- 1: partially correct or generic;
- 2: correct;
- 3: correct and well justified.

Suggested criteria include objective of the form, entities, fields, business rules, validations, exceptions and alerts, relations with other documents, ambiguities, technical usefulness, and risk of invented rules. Reviewers should be blind to condition A or B.

References

- [1] Klaus Pohl. *Requirements Engineering: Fundamentals, Principles, and Techniques*. Springer, 2010.
- [2] Ian Sommerville. *Software Engineering*. 9th edition. Addison-Wesley, 2011.
- [3] Karl Wiegers and Joy Beatty. *Software Requirements*. 3rd edition. Microsoft Press, 2013.
- [4] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. Large language models for software engineering: A systematic literature review. *ACM Transactions on Software Engineering and Methodology*, 2024.
- [5] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. A prompt pattern catalog to enhance prompt engineering with ChatGPT. *arXiv preprint arXiv:2302.11382*, 2023.
- [6] Lewis R. Goldberg. The development of markers for the Big-Five factor structure. *Psychological Assessment*, 4(1):26–42, 1992.
- [7] Paul T. Costa and Robert R. McCrae. *Revised NEO Personality Inventory (NEO-PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual*. Psychological Assessment Resources, 1992.
- [8] Lewis R. Goldberg. A broad-bandwidth, public-domain, personality inventory measuring the lower-level facets of several five-factor models. In *Personality Psychology in Europe*, volume 7, pages 7–28, 1999.
- [9] John A. Johnson. Measuring thirty facets of the Five Factor Model with a 120-item public domain inventory. *Journal of Research in Personality*, 51:78–89, 2014.